RAJALAKSHMI ENGINEERING COLLEGE

Dhanusheya G 251801053-2025-AIDS _ C2

Started on	Thursday, 17 September 2026, 8:05 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:18 AM
Time taken	12 mins 49 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    // No counter increment needed for scanf per the instructions
    if (scanf("%d", &n) != 1) {
        return 1;
    }

    int count = 0;

    // Initializations (Assignments count as 1 step each)
    int i = 1; count++;
    int s = 1; count++;

    while (s <= n) {
        count++; // Incremented for a successful while-condition check
        i++;     count++; // Incremented for body statement 1
        s += i;  count++; // Incremented for body statement 2
    }
    count++; // Incremented for the final while-condition check that evaluates to false

    // Output the total step count
    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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